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China 2026

From Containers to Agents: The Next Cloud Native Abstraction

Di Xu (@dixudx, Xiaohongshu)

Yuqi Huang (Xiaohongshu)



A New Unit of Work



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Every major computing shift creates a new unit of work.

Era	Primary Abstraction	What It Changed
Virtualization	Virtual Machine	Infrastructure became programmable.
Cloud Native	Container	Applications became portable.
Kubernetes	Workload	Distributed systems became operable.
AI Native	Agent	Software can reason and take action.

*“Containers are still the execution unit.
Agents are becoming a new unit of autonomous work.”*

What Containers and Kubernetes Solved



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- Containers changed how we package software.

Portable runtimes / Consistent environments / Resource isolation / Repeatable deployment / Declarative configuration

Build once → Run anywhere

- Kubernetes changed how we operate software.

Desired State



Declarative APIs



Controllers reconcile reality



Scalable, resilient, observable workloads

Agents Are Different



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- An agent is not just a service with an LLM inside.

*A traditional service follows a **predictable** request-response flow.*

An agent receives a goal and decides how to act.

- An agent can be

Stateful / Long-running / Non-deterministic / Tool-connected / Delegated with authority

“An agent does not only respond. An agent can act.”

A Healthy Runtime Can Still Be Unsafe



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- Traditional infrastructure asks: “Is the workload healthy?”

Is the Pod running? Is the API available?

Is latency within the SLO? Is the error rate acceptable?

Is CPU or GPU utilization normal?

- Agent infrastructure must also ask: “Did the agent behave correctly?”

Pod: Running ✓

API: 200 OK ✓

Latency: 800ms ✓

Tool Call: Successful ✓

Customer Outcome: Wrong ✗

“Runtime health is necessary—but not sufficient.”

The New Operational Problem



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Operating autonomous software requires new answers.

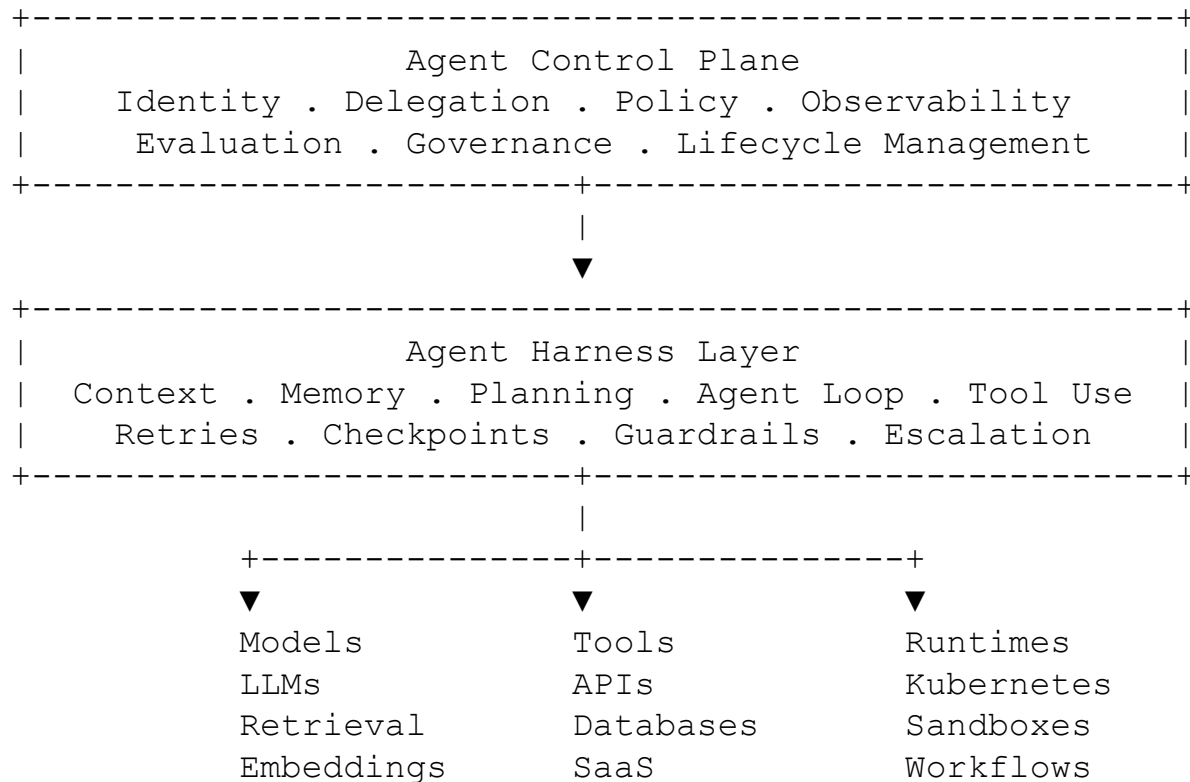
Question	Why It Matters
Who is this agent?	Identity and auditability
Who authorized it to act?	Delegation and accountability
What can it access?	Data and tool permissions
What actions can it take?	Safety and policy enforcement
Why did it make this decision?	Debugging and observability
Did it achieve the right outcome?	Evaluation and reliability
When should a human intervene?	Governance and approval

The Agent Infrastructure



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“Agent infrastructure is not a replacement for Kubernetes. It is an operational layer for autonomous behavior.”



Four Core Primitives



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1. Identity and Delegation

Every action must be attributable to an agent and its delegator.

2. Policy-Driven Tool Access

Agents need least-privilege access—not broad API keys.

3. Decision-Aware Observability

We need to observe decisions and outcomes—not only latency and errors.

4. Continuous Evaluation

Evaluation is the reconciliation loop for agent behavior.

Observe Decisions, Not Just Requests



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- Traditional observability follows requests.

Request → Service → Database → Response

- Agent behavior is not static. Agent observability must follow decisions and actions.

User Goal

→ *Retrieved Context*

→ *Model Decision*

→ *Selected Tool*

→ *Tool Arguments*

→ *Tool Result*

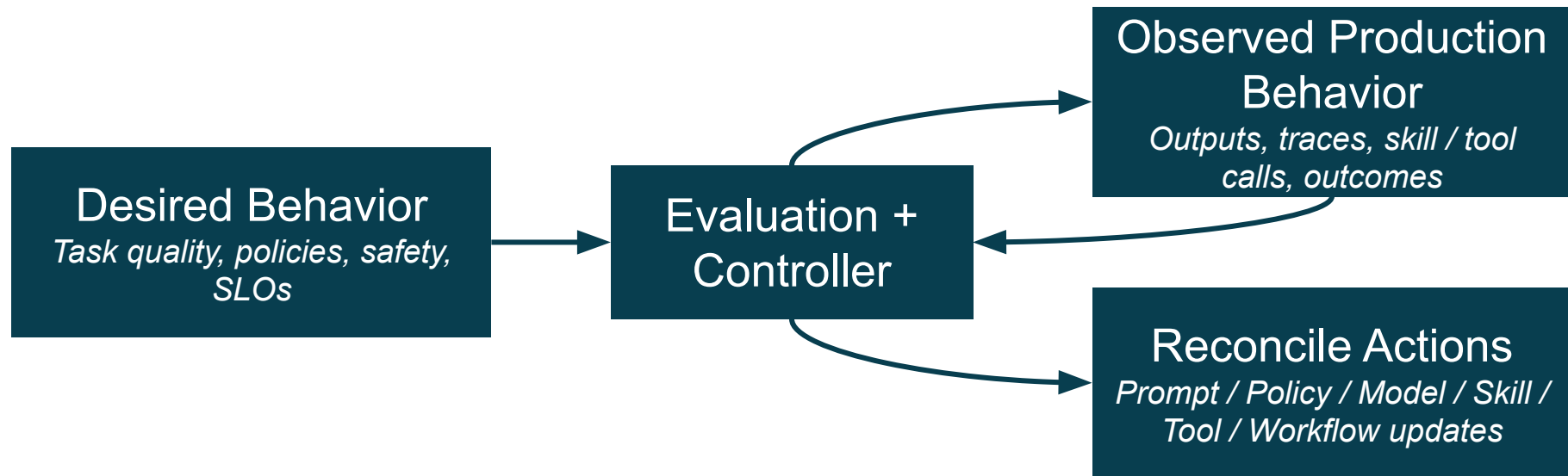
→ *Final Action*

→ *Outcome and Feedback*

Evaluation Is the Reconciliation Loop



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Cloud Native Gives Us the Right Mental Model



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We can reuse cloud native lessons, but agents require new declarative boundaries.

Cloud Native Primitive	Agent Infrastructure Equivalent
Deployment / Service / Pod	Agent runtime configuration / definition
ServiceAccount	Agent identity
RBAC	Tool and action authorization
NetworkPolicy	Tool connectivity boundaries
Admission Control	Policy checks before action
OpenTelemetry	Decision and action traces
Controller Reconciliation	Continuous evaluation and correction

The Opportunity: Open Agent Infrastructure



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- *Cloud native succeeded because its primitives were open, declarative, and composable.*
- Build open, composable primitives for fast-moving agent ecosystem:
 - Agent identity*
 - Delegation*
 - Tool access*
 - Policy enforcement*
 - Observability*
 - Evaluation*
 - Governance*

Closing



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Containers made software portable.

Kubernetes made software operable.

Agent infrastructure must make autonomous software **trustworthy**.

“Operate agents as first-class cloud native workloads.”



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Thanks for Listening

